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[1]: # =====
# EXPERIMENT 3(A)
# UNIVARIATE ANALYSIS: STATISTICAL ANALYSIS USING
# DIABETES DATASETS
# =====

import pandas as pd
import numpy as np
from scipy.stats import mode, skew, kurtosis

print("=" * 75)
print("EXPERIMENT 3(A) - UNIVARIATE ANALYSIS")
print("=" * 75)

# =====
# 1. CREATE SAMPLE UCI DIABETES DATASET
# =====

uci_data = {
    "Glucose": [
        90, 162, 184, 119, 175,
        137, 145, 111, 120, 130
    ],

    "BloodPressure": [
        72, 61, 76, 83, 101,
        82, 85, 63, 80, 78
    ],

    "SkinThickness": [
        16, 34, 25, 20, 30,
        29, 30, 39, 25, 35
    ],

    "Insulin": [
        264, 143, 128, 230, 63,
        146, 142, 290, 120, 180
    ],

    "BMI": [
        22.273511,
        41.814986,
        36.309833,
        27.258685,
        41.493982,
        30.990932,
        30.298310,
        18.528511,
        29.500000,
        31.200000
    ]
}
```



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# =====
# 2. CREATE SAMPLE PIMA INDIANS DIABETES DATASET
# =====

pima_data = {
    "Glucose": [
        81, 172, 90, 187, 133,
        136, 135, 122, 140, 155
    ],

    "BloodPressure": [
        97, 50, 84, 94, 94,
        82, 82, 52, 80, 90
    ],

    "SkinThickness": [
        15, 24, 40, 30, 46,
        29, 30, 17, 25, 35
    ],

    "Insulin": [
        289, 229, 136, 288, 23,
        148, 152, 160, 125, 200
    ],

    "BMI": [
        27.062801,
        18.588741,
        32.968461,
        38.217097,
        37.236422,
        32.479119,
        32.634410,
        18.588741,
        30.500000,
        34.200000
    ],

    "DiabetesPedigreeFunction": [
        0.797425,
        0.652107,
        0.952396,
        1.036946,
        1.494192,
        1.239018,
        1.213841,
        0.121055,
        0.500000,
        0.700000
    ],
}
```

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)

print(
    "The analysis reveals differences in central tendency, "
    "dispersion, and distribution between the two datasets."
)

print("=" * 75)
print("EXPERIMENT 3(A) COMPLETED SUCCESSFULLY")
print("=" * 75)
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EXPERIMENT 3(A) - UNIVARIATE ANALYSIS
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UCI DIABETES DATASET SAMPLE
=====
  Glucose  BloodPressure  SkinThickness  Insulin  BMI  \
0      90           72           16      264  22.273511
1     162           61           34      143  41.814986
2     184           76           25      128  36.309833
3     119           83           20      230  27.258685
4     175          101           30       63  41.493982

  DiabetesPedigreeFunction  Age  Outcome
0              1.162896    79         0
1              2.158993    35         0
2              1.411238    31         0
3              1.202206    28         1
4              0.214888    59         1

=====
PIMA INDIANS DIABETES DATASET SAMPLE
=====
  Glucose  BloodPressure  SkinThickness  Insulin  BMI  \
0      81           97           15      289  27.062801
1     172           50           24      229  18.588741
2      90           84           40      136  32.968461
3     187           94           30      288  38.217097
4     133           94           46       23  37.236422

  DiabetesPedigreeFunction  Age  Outcome
0              0.797425    62         1
1              0.652107    21         1
2              0.952396    29         0
3              1.036946    35         1
4              1.494192    62         0

=====
UCI DIABETES DATASET STATISTICS
=====
              Mean      Median      Mode  Variance  \
Glucose      137.300000  133.500000  90.000000  876.455556
BloodPressure  78.100000  79.000000  61.000000  130.766667
```

===== PIMA INDIANS DIABETES DATASET STATISTICS =====

	Mean	Median	Mode	Variance \
Glucose	135.100000	135.500000	81.000000	1068.100000
BloodPressure	80.500000	83.000000	82.000000	276.277778
SkinThickness	29.100000	29.500000	30.000000	92.100000
Insulin	175.800000	160.000000	23.000000	6414.622222
BMI	30.247579	32.556764	18.588741	47.569138
DiabetesPedigreeFunction	0.870698	0.874910	0.121055	0.162015
Age	43.500000	47.500000	50.000000	222.055556

	Standard Deviation	Skewness	Kurtosis
Glucose	32.681799	-0.182157	-0.619271
BloodPressure	16.621606	-1.050838	-0.292485
SkinThickness	9.596874	0.205697	-0.688429
Insulin	80.091337	-0.175390	-0.340937
BMI	6.897038	-0.802390	-0.608594
DiabetesPedigreeFunction	0.402511	-0.269640	-0.544643
Age	14.901529	-0.200903	-1.360445

===== MEAN COMPARISON =====

	UCI Diabetes	Pima Indians
Glucose	137.3000	135.1000
BloodPressure	78.1000	80.5000
SkinThickness	28.3000	29.1000
Insulin	170.6000	175.8000
BMI	30.9669	30.2476
DiabetesPedigreeFunction	0.9921	0.8707
Age	50.7000	43.5000

===== STANDARD DEVIATION COMPARISON =====

	UCI Diabetes	Pima Indians
Glucose	29.6050	32.6818
BloodPressure	11.4353	16.6216
SkinThickness	6.9929	9.5969
Insulin	70.5017	80.0913
BMI	7.4814	6.8970
DiabetesPedigreeFunction	0.6352	0.4025
Age	16.6002	14.9015

===== RESULT =====

The univariate analysis of the UCI Diabetes and Pima Indians Diabetes datasets was successfully performed. Mean, median, mode, variance, standard deviation, skewness, and kurtosis were calculated for the selected numerical variables. The analysis reveals differences in central tendency, dispersion, and distribution between the two datasets.

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EXPERIMENT 3(A) COMPLETED SUCCESSFULLY

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